

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Semester:- III

Mid-Term Examination September-2026

Course : B.Tech

Subject Code :- CSE(DS)-3001/ CSE(CS)-3001

Max. Marks: 20

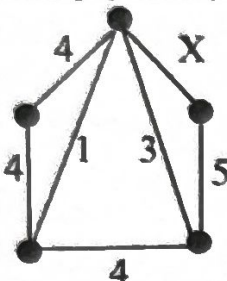
Subject Name: Design and Analysis of Algorithms

Time : 90 Minutes

NOTE:

1. All the questions are compulsory.
2. While solving write question number and its sub part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.

1.A	Consider the following recurrence relation: $T(n) = 2T(n-1)+n2^n$ for $n>0$, $T(0)=1$. Calculate the complexity of the above relation.	3												
1.B	$A(n)$ { if($n\leq 2$) return 2; else return $A(\sqrt{n})$; } What is the time complexity of the above function.	2												
2.	Generate a Huffman coding scheme for a file containing the following characters and their given frequencies. Show the total number of bits required for encoding a file of 60,000 bits using the generated code. <table><tr><td>Character</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>Frequency</td><td>24</td><td>8</td><td>10</td><td>12</td><td>6</td></tr></table>	Character	A	B	C	D	E	Frequency	24	8	10	12	6	5
Character	A	B	C	D	E									
Frequency	24	8	10	12	6									
3.A	You are given a max heap stored in an array. The following function is used to delete a node from the heap: void deleteRoot(int array[], int num) { int i; for (i = 0; i < size; i++) { if (num == array[i]) break;} swap(&array[i], &array[size - 1]);	4												

	<pre>size = 1; for (int i = size / 2 - 1; i >= 0; i--) {heapify(array, size, i);}</pre> (i) Determine the time complexity of the above algorithm and provide a detailed mathematical justification. (ii) Write an algorithm that can reduce the time complexity to delete a node from the heap.	
3.B	 Consider the above undirected graph G. Choose a value for X that will maximize the no of minimum weight spanning tree of G.	1
4.	Consider two strings A = "qpqrpr" and B = "pqprqrp". Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B. What is the value of x and y [Solve using a dynamic programming approach; write the recurrence relations required to solve the above problem.]	5